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Statistical Models

Week 8: Bayesian statistics: the idea, and conjugate priors

This week, will start with the third big topic: Bayesian statistics.

Lectures and exercises will be mainly based on the book BJ Reich, SK Ghosh (2019): Bayesian Statistical Methods.

The book starts with a review of probability background (Section 1.1) and of the Bayes rule for calculating conditional probabilities / conditional densities (Section 1.2). You can use these introductory sections for reference :-)

I will very briefly recall the Bayes rule in the first lecture on Wednesday. Then, I will introduce the basic ideas of Bayesian statistics.

On Friday, we will learn about conjugate priors: prior distributions that match well with the likelihood, such that the form of the posterior is analytically known. Then we will look at the distinction of informative and uninformative priors, and "improper" priors.

If time allows, we have a look at the paper by Mr.Bayes and Mr.Price (1763): "An Essay towards Solving a Problem in the Doctrine of Chances", *Philosophical Transactions*, 1763, Vol. 53, pp. 370-418, where they introduced the idea of sequentially updating one's believe by experiments.

Relevant Readings

- Probability theory background (review): R&G, Section 1.1, 1.2
- Basic ideas of Bayesian statistics: R&G, Chapter 1: mainly 1.3-1.5
- Conjugate priors; RG, Sections 2.1.1-2.1.3 and 2.1.5
- Informative and uninformative priors, and "improper" priors: R&G, Section 2.2

Lecture Material

- Link to onenotes on Google drive
- Example for calculating and plotting the posterior: Reich & Ghosh, Exercise 1.14

Exercises 8

Problem 8.1: Conditional probability of species detection

Reich & Ghosh, Chapter 1, Exercise 10
Let $X_1 \sim \text{Bernoulli}(\theta)$ be the indicator that a tree species occupies a forest and $\theta \in [0,1]$ denote the prior occupancy probability.
The researcher gathers a sample of n trees from the forest and X_2 belong to the species of interest. The model for the data is $X_2 \mid X_1 \sim \text{Binomial}(n, \lambda X_1)$ where $\lambda \in [0,1]$ is the probability of detecting the species given it is present.
Give expressions in terms of n, θ and λ for the following joint, marginal and conditional probabilities:

a. $\text{P}(X_1 = X_2 = 0)$.
b. $\text{P}(X_1 = 0)$.

c. $P(X_2 = 0)$.
d. $P(X_1 = 0 \mid X_2 = 0)$.
e. $P(X_2 = 0 \mid X_1 = 0)$.
f. $P(X_1 = 0 \mid X_2 = 1)$.
g. $P(X_2 = 0 \mid X_1 = 1)$.
h. Provide intuition for how (d)-(g) change with n, θ and λ.
i. Assuming $\theta = 0.5$, $\lambda = 0.1$, and $X_2 = 0$, how large must n be before we can conclude with 95% confidence that the species does not occupy the forest?

Problem 8.2: Bayesian classification of vegetation images

Reich & Ghosh, Chapter 1, Exercise 13
The normalized difference vegetation index (NDVI) is commonly used to classify land cover using remote sensing data. Hypothetically, say that NDVI follows a Beta$(25, 10)$ distribution for pixels in a rain forest, and a Beta$(10, 15)$ distribution for pixels in a deforested area now used for agriculture. Assuming about 10% of the rain forest has been deforested, your objective is to build a rule to classify individual pixels as deforested based on their NDVI.

a. Plot the PDF of NDVI for forested and deforested pixels, and the marginal distribution of NDVI averaging over categories.

b. Give an expression for the probability that a pixel is deforested given its NDVI value, and plot this probability by NDVI.

c. You will classify a pixel as deforested if you are at least 90% sure it is deforested. Following this rule, give the range of NDVI that will lead to a pixel being classified as deforested.

Problem 8.3: Bayesian prediction of seedling survival

Reich & Ghosh, Chapter 1, Exercise 15
Last spring your lab planted ten seedlings and two survived the winter. Let θ be the probability that a seedling survives the winter.

a. Assuming a uniform prior distribution for θ, compute its posterior mean and standard deviation.

b. Assuming the same prior as in (a), compute and compare the equal-tailed and highest density 95% posterior credible intervals.

c. If you plant another 10 seedlings next year, what is the posterior predictive probability that at least one will survive the winter?

Problem 8.4: Bayesian analysis of sports data: good or bad season?

see Reich & Ghosh, Chapter 1, Exercise 17
The table below shows free throw results in pressure situations (called "clutch" https://stats.nba.com/) for ten National Basketball Association players (those that received the most votes for the Most Valuable Player Award). The column "Overall proportion" stands for a long term average of success rate before 2016, while "Clutch makes" and "Clutch attempts" refer to the 2016/2017 season.
When you calculate the success rate in the 2016/17 season, it will deviate in most cases from the long term result. Does this mean the player's capabilities have changed?

Player	Clutch makes	Clutch attempts	Overall proportion
Russell Westbrook	64	75	0.845
James Harden	72	95	0.847
Kawhi Leonard	55	63	0.880
LeBron James	27	39	0.674
Isaiah Thomas	75	83	0.909
Stephen Curry	24	26	0.898
Giannis Antetokounmpo	28	41	0.770
John Wall	66	82	0.801
Anthony Davis	40	54	0.802
Kevin Durant	13	16	0.875

Analyze the clutch data for each player separately using Bayesian methods. Assume a prior throughout this problem.

a. Describe your model for studying the clutch success probability including the likelihood and prior.

b. Plot the posteriors of the clutch success probabilities.

c. Summarize the posteriors in a table, including posterior median and limits of a 95% credible interval.

d. Do you find evidence that the current percentage of makes differs from the overall percentage for any of the players?

e. Are the results sensitive to your prior? That is, do small changes in the prior lead to substantial changes in the posterior?

Remarks:
For your convenience, you can download the data as file `NBA-clutches.csv` or find it in the `data` folder on Google Drive.

Hints

- A distribution of the form $f(x) \propto x^{\alpha-1}(1-x)^{\beta-1}$ is a Beta(α, β) distribution.
- In (b) you should find that the posteriors are Beta distributions. Then you can use R's built in functions to plot the posteriors and calculate probabilities and quantiles.
- In (e) it makes sense to use a Beta distribution as prior. Choose parameters such that it is not too different from the uniform.

Problem 8.5: Bayesian classification of authors

Reich & Ghosh, Chapter 1, Exercise 18. Very American background ;-)
In the early twentieth century, it was generally agreed that Hamilton and Madison (ignore Jay for now) wrote 51 and 14 Federalist Papers, respectively. There was dispute over how to attribute 12 other papers between these two authors. In the 51 papers attributed to Hamilton the word "upon" was used 3.24 times per 1,000 words, compared to 0.23 times per 1,000 words in the 14 papers attributed to Madison (for historical perspective on this problem, see Mosteller and Wallace, 1963[a])

a. If the word "upon" is used three times in a disputed text of length 1,000 words and we assume the prior probability 0.5, what is the posterior probability the paper was written by Hamilton?

b. Give one assumption you are making in (a) that is likely unreasonable. Justify your answer.

c. In (a), if we changed the number of instances of "upon" to one, do you expect the posterior probability to increase, decrease or stay the same? Why?

d. In (a), if we changed the text length to 10,000 words and number of instances of "upon" to 30, do you expect the posterior probability to increase, decrease or stay the same? Why?

e. Let Y be the number of observed number of instances of "upon" in 1,000 words. Compute the posterior probability the paper was written by Hamilton for each $Y \in \{0, 1, \ldots, 20\}$ plot these posterior probabilities versus Y and give a rule for the number of instances of "upon" needed before the paper should be attributed to Hamilton.

[4] Frederick Mosteller and David L Wallace (1963): Inference in an authorship problem: A comparative study of discrimination methods applied to the authorship of the disputed Federalist Papers. *Journal of the American Statistical Association*, **58**(302):275-309.

General hints and remarks:

Discrete priors to select between likelihoods

In Exercise 8.2, we want to decide between different likelihoods, corresponding to different reasons for these likelihoods (in these exercises, we have only two options to choose between). Then we assign a probability to each reason. This means the prior is discrete, and the posterior assigns a new probability to the reasons. The denominator in the formula for the posterior distribution (1.36 in Ghosh) is then a sum.

R to solve a nonlinear equation

If you need to solve an equation of the form $g(x) = 0$ that looks hopeless, remember that function `uniroot` can be helpful (e.g. Exercise 8.1 c)